

Lab #3

Due at the start of your next lab period

Objectives

Use repetition statements

Solve a numerical engineering problem

Format output

Requirements

Write a C program that computes the height and velocity of a falling object, including the effects of atmospheric drag.

Background

From simple physics, we know that $F = ma$. For a free-falling object with no drag, the object will accelerate due to gravity. We have the following functions of time for velocity and displacement

$$\begin{aligned} v &= gt \\ d &= \frac{1}{2}gt^2 \end{aligned}$$

We could also solve the second equation to find the time that it takes to cover a given distance.

When nonlinear drag and altitude effects are included, *we can no longer use the preceding equations* to solve the problem. However, we can apply the force-balance equations repeatedly over small time intervals, solving these equations at *each point in time*.

$$\begin{aligned} F_g &= mg \\ F_d &= -\frac{1}{2}C\rho Av^2 \\ F_{net} &= F_g + F_d \\ F_{net} &= ma \end{aligned}$$

where C is the drag coefficient, ρ is the density of the air, A is the cross-sectional area, and v is the velocity at the current time step. We can then use the net force to compute the acceleration and use that value to update the velocity and displacement at that point in time.

As the object falls, it will increase in velocity until the force due to gravity is balanced by the drag force. It has reached its *terminal velocity*.

We will assume that drag has a quadratic dependency on velocity, which is appropriate for large objects.

To make this lab a little more interesting, we will compute the air density as a function of altitude! We have (from Wikipedia):

p_0	=	101.325	kPa	sea-level standard atmospheric pressure
T_0	=	288.15	K	sea level standard temperature
g	=	9.80665	m/s ²	earth-surface gravitational acceleration
L	=	0.0065	K/m	temperature lapse rate
R	=	8.31447	J/(mol·K)	ideal (universal) gas constant
M	=	0.0289644	kg/mol	molar mass of dry air

We can then estimate the temperature at altitude using

$$T = T_0 - Lh$$

and then the pressure at altitude is

$$p = p_0 \left(1 - \frac{Lh}{T_0} \right)^{\frac{gM}{RL}}$$

Now, we can compute the density as

$$\rho = \frac{pM}{RT}$$

Fortunately (for you), I have provided a useful function in the file `density.c`

Although you probably haven't done this in math yet: we are solving a *nonlinear* second-order ordinary differential equation using numerical methods. The technique that we are using is called Euler's method, which we will look at in the next course.

Requirements (continued)

Your program must:

- 1) Prompt for the drop height, mass, drag coefficient, cross-sectional area, and time-step size
- 2) Compute and display the time, height, and velocity at each time step. Format as shown below, so be sure to use field-width specifiers in your `printf` statements.
- 3) Stop when the height of the object becomes negative
- 4) Add a line of code at the end of your program so that you can compare your terminal velocity with the simple formula at:

<http://hyperphysics.phy-astr.gsu.edu/hbase/airfri2.html>

Hints and Tips

- 1) The solution is iterative, i.e., it repeatedly uses the force balance equations to compute the changing acceleration, velocity, and displacement.

Evaluation

A working, well-documented program to the instructor	9/10
Use for to limit the maximum number of iterations to 100,000	1/10

Sample Run

```
Enter the height in metres: 1500
Enter the mass in kg: 80
Enter the drag coefficient: 0.5
Enter the cross-sectional area: 0.4
Enter the time step size: 0.1
```

Time	Height	Velocity
0.00	1500.0	0.00
0.10	1499.9	0.98
0.20	1499.7	1.96
0.30	1499.4	2.94
0.40	1499.0	3.92
0.50	1498.5	4.90

0.60	1497.9	5.88
0.70	1497.3	6.85
0.80	1496.5	7.83
0.90	1495.6	8.80
1.00	1494.6	9.77
1.10	1493.5	10.74
1.20	1492.4	11.70
1.30	1491.1	12.67
1.40	1489.7	13.63
1.50	1488.3	14.58
1.60	1486.7	15.53
1.70	1485.1	16.48
1.80	1483.3	17.43
1.90	1481.5	18.37
2.00	1479.6	19.30

Many missing lines

22.00	144.7	80.97
22.10	136.6	80.96
22.20	128.5	80.95
22.30	120.4	80.94
22.40	112.3	80.93
22.50	104.2	80.92
22.60	96.1	80.90
22.70	88.0	80.89
22.80	79.9	80.88
22.90	71.9	80.86
23.00	63.8	80.85
23.10	55.7	80.84
23.20	47.6	80.82
23.30	39.5	80.81
23.40	31.4	80.79
23.50	23.4	80.78
23.60	15.3	80.76
23.70	7.2	80.74

Theoretical terminal velocity at sea level is 80.0 m/s

